

Brewery Plat-Eau: Project Answers for Beer Production Optimisation

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1. INTRODUCTION

BREWERY Plat-Eau is an optimisation project about choosing how to produce beer at minimum cost. The model decides how much barley, malt, malt extract, hops, and yeast to use for a requested beer volume and colour. Later tasks also require minimum ingredient quality.

The continuous decision variables are ingredient quantities. Two binary variables represent whether the malting and mashing processes are activated, because these processes have fixed costs. The report is organised as a direct answer to the assignment tasks: first the common brewing assumptions and optimisation model are defined, and then Tasks 1–7 are answered in order.

2. PROBLEM BACKGROUND AND BREWING ASSUMPTIONS

Only the brewing assumptions needed for the optimisation model are used. The fermentable material can be supplied through three routes. Raw barley is first converted to malt and then mashed. Purchased malt skips the malting step but still requires mashing. Malt extract is already in malt-extract-equivalent form and can be used directly.

The final beer volume is related to wort volume by

$$V_{\text{beer}} = 0.95V_{\text{wort}}, \quad (1)$$

so that a requested beer volume implies

$$V_{\text{wort}} = \frac{V_{\text{beer}}}{0.95}. \quad (2)$$

The required malt-extract-equivalent mass is

$$\frac{M_{\text{tot}}}{V_{\text{wort}}} = 2.9(SG - 1), \quad SG = 1.050. \quad (3)$$

Therefore

$$M_{\text{tot}} = 2.9(1.050 - 1)V_{\text{wort}}. \quad (4)$$

Barley and malt conversion are

$$1 \text{ kg barley} = 0.75 \text{ kg malt}, \quad (5)$$

$$1 \text{ kg malt} = 0.8 \text{ kg malt extract equivalent}. \quad (6)$$

Consequently,

$$1 \text{ kg barley} = 0.6 \text{ kg malt extract equivalent}. \quad (7)$$

Hops are required in proportion to wort volume, and yeast is required in proportion to final beer volume:

$$M_{\text{hops}} = 0.0013V_{\text{wort}}, \quad (8)$$

$$M_{\text{yeast}} = 0.00075V_{\text{beer}}. \quad (9)$$

Beer colour is measured by EBC. Each beer colour in the assignment is represented by a lower and, except for black beer, an upper EBC bound. Quality is treated as a weighted average inside each ingredient group. Therefore, quality requirements can be written as linear weighted-average constraints once the total amount of each group is fixed.

All ingredient costs, EBC values, qualities, and process costs are read from `dataset_EN.txt`. The Python code keeps the dataset as the single input source for the numerical tasks.

3. MODEL SETUP

A. Decision Variables

Let B, M, E, H , and Y denote the sets of barley, malt, malt-extract, hop, and yeast alternatives. The continuous decision variables are

$$\begin{aligned} b_i &\geq 0 & i \in B, & \text{kg of barley } i, \\ m_i &\geq 0 & i \in M, & \text{kg of malt } i, \\ e_i &\geq 0 & i \in E, & \text{kg extract-equivalent of extract } i, \\ h_i &\geq 0 & i \in H, & \text{kg of hops } i, \\ y_i &\geq 0 & i \in Y, & \text{kg of yeast } i. \end{aligned}$$

The binary process variables are

$$\begin{aligned} z_{\text{malt}} &\in \{0, 1\}, & 1 \text{ if malting is used,} \\ z_{\text{mash}} &\in \{0, 1\}, & 1 \text{ if mashing is used.} \end{aligned}$$

For compact notation, the extract-equivalent coefficients are

$$a^B = 0.6, \quad a^M = 0.8, \quad a^E = 1. \quad (10)$$

B. Objective Function

The cost-minimisation objective includes ingredient costs, variable process costs, and fixed process costs:

$$\begin{aligned} \min C = & \sum_{i \in B} c_i^B b_i + \sum_{i \in M} c_i^M m_i + \sum_{i \in E} c_i^E e_i \\ & + \sum_{i \in H} c_i^H h_i + \sum_{i \in Y} c_i^Y y_i \\ & + v_{\text{malt}} \sum_{i \in B} b_i + v_{\text{mash}} \left(0.75 \sum_{i \in B} b_i + \sum_{i \in M} m_i \right) \\ & + F_{\text{malt}} z_{\text{malt}} + F_{\text{mash}} z_{\text{mash}}. \end{aligned} \quad (11)$$

Here F denotes fixed process costs and v denotes variable process costs from the dataset.

C. Volume Constraints

For a requested final beer volume V_{beer} , the required wort volume and malt-extract-equivalent mass are

$$V_{\text{wort}} = \frac{V_{\text{beer}}}{0.95}, \quad (12)$$

$$M_{\text{tot}} = 2.9(1.050 - 1)V_{\text{wort}}. \quad (13)$$

The fermentable-source balance is

$$0.6 \sum_{i \in B} b_i + 0.8 \sum_{i \in M} m_i + \sum_{i \in E} e_i = M_{\text{tot}}. \quad (14)$$

The hop and yeast balances are

$$\sum_{i \in H} h_i = 0.0013V_{\text{wort}}, \quad (15)$$

$$\sum_{i \in Y} y_i = 0.00075V_{\text{beer}}. \quad (16)$$

D. Colour Constraints

The EBC formula is a weighted average over fermentable sources. Barley and malt are first converted to malt-extract-equivalent mass before the EBC average is formed. Define

$$\gamma = \frac{SG - 1}{0.0344} \quad (17)$$

and the weighted colour contribution

$$\begin{aligned} \Phi_E &= \sum_{i \in B} E_i^B (0.6b_i) + \sum_{i \in M} E_i^M (0.8m_i) \\ &+ \sum_{i \in E} E_i^E e_i. \end{aligned} \quad (18)$$

Then

$$\text{EBC} = \gamma \frac{\Phi_E}{M_{\text{tot}}}. \quad (19)$$

For a colour interval $[\text{EBC}_{\min}, \text{EBC}_{\max}]$, this becomes the pair of linear bounds

$$\gamma \Phi_E \geq \text{EBC}_{\min} M_{\text{tot}}, \quad (20)$$

$$\gamma \Phi_E \leq \text{EBC}_{\max} M_{\text{tot}}. \quad (21)$$

The assignment defines black beer as $\text{EBC} > 120$. Since strict inequalities cannot be represented directly by an LP solver, the implementation uses

$$\text{EBC} \geq 120 + \epsilon_{\text{EBC}}, \quad \epsilon_{\text{EBC}} = 10^{-6}, \quad (22)$$

and omits the upper EBC constraint when $\text{EBC}_{\max} = \infty$.

E. Quality Constraints

Quality requirements are weighted averages rewritten as linear inequalities. For fermentable sources,

$$\sum_{i \in B} Q_i^B (0.6b_i) + \sum_{i \in M} Q_i^M (0.8m_i) + \sum_{i \in E} Q_i^E e_i \geq Q_{\min}^S M_{\text{tot}}. \quad (23)$$

For hops and yeast,

$$\sum_{i \in H} Q_i^H h_i \geq Q_{\min}^H \sum_{i \in H} h_i, \quad (24)$$

$$\sum_{i \in Y} Q_i^Y y_i \geq Q_{\min}^Y \sum_{i \in Y} y_i. \quad (25)$$

F. Process Constraints

Big-M constraints link ingredient use to the fixed-cost process variables. Barley requires malting, barley and purchased malt require mashing, and malting implies mashing:

$$\sum_{i \in B} b_i \leq M_{\text{malt}} z_{\text{malt}}, \quad (26)$$

$$0.75 \sum_{i \in B} b_i + \sum_{i \in M} m_i \leq M_{\text{mash}} z_{\text{mash}}, \quad (27)$$

$$z_{\text{malt}} \leq z_{\text{mash}}. \quad (28)$$

The implementation uses

$$M_{\text{malt}} = \frac{M_{\text{tot}}}{0.6}, \quad M_{\text{mash}} = \frac{M_{\text{tot}}}{0.8}, \quad (29)$$

which are large enough to allow all-barley or all-malt solutions.

G. LP Relaxation and Brute-Force Solution

The LP relaxation replaces the binary conditions

$$z_{\text{malt}}, z_{\text{mash}} \in \{0, 1\} \quad (30)$$

with continuous bounds

$$0 \leq z_{\text{malt}}, z_{\text{mash}} \leq 1. \quad (31)$$

The relaxation gives a lower bound for cost minimisation, but fractional process variables are not valid production decisions.

The original MILP is solved by brute force over the valid binary process choices

$$(z_{\text{malt}}, z_{\text{mash}}) \in \{(0, 0), (0, 1), (1, 1)\}. \quad (32)$$

For each fixed process choice, one LP is solved using `scipy.optimize.linprog`. Since the model has only two binary variables, this enumeration is exact and does not require a dedicated MILP solver.

4. TASK 1: FIRST ANALYSIS

This section answers assignment Task 3.1 by first converting a fixed barley amount to beer volume and then comparing cost as a function of beer volume for the three fermentable-source routes.

A. Beer Volume from 25.44 kg Barley

Using (7) and (3), the beer volume obtainable from a barley mass M_B is

$$V_{\text{beer}} = 0.95 \frac{0.6M_B}{2.9(SG - 1)}. \quad (33)$$

For $M_B = 25.44$ kg, the extract-equivalent mass is 15.264 kg, the wort volume is 105.269 L, and the final beer volume is 100.006 L.

TABLE I
TASK 1 BARLEY-TO-BEER CALCULATION

Quantity	Expression	Value
Barley mass	M_B	25.44 kg
Extract equivalent	$0.6M_B$	15.264 kg
Wort volume	$0.6M_B/[2.9(SG - 1)]$	105.269 L
Beer volume	$0.95V_{\text{wort}}$	100.006 L

TABLE II
TASK 3 PROBLEM SETUP

Parameter	Value
Beer volume	5000 L
Colour	amber
EBC bounds	20 to 29
Quality constraints	not active

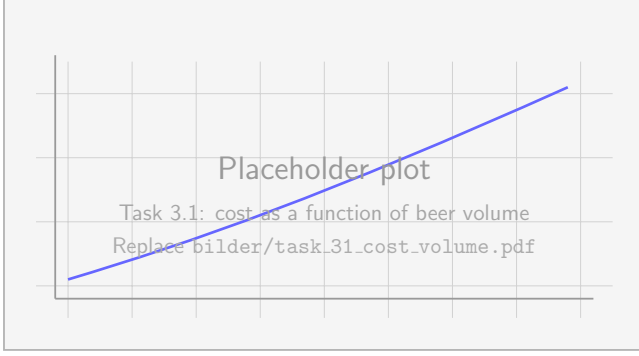


Fig. 1. Task 1 cost as a function of beer volume. Replace this placeholder with the plot exported from `task_31.ipynb`.

B. Cost Functions for Separate Malt-Source Routes

The first cost analysis uses the cheapest available barley, malt, malt extract, hops, and yeast from the dataset. With V denoting final beer volume in litres, the separate-route cost functions from `task_31.ipynb` are

$$C_{\text{barley}}(V) = 3482.000 + 0.397529V, \quad (34)$$

$$C_{\text{malt}}(V) = 1072.000 + 0.661836V, \quad (35)$$

$$C_{\text{extract}}(V) = 1.338757V. \quad (36)$$

The barley route has the largest fixed cost because both malting and mashing are activated. The malt route has only the mashing fixed cost. The malt-extract route has no process fixed cost but has the largest variable cost per litre.

C. Combined Malt-Source Solution

When all three source routes can be chosen, the combined minimum-cost curve is the lower envelope of the three separate cost curves. For small volumes, malt extract is cheapest because it avoids fixed process costs. For medium volumes, purchased malt becomes cheaper. For large volumes, raw barley becomes attractive because its variable cost per litre is lowest, even though it has the highest fixed process cost.

5. TASK 2: LP AND MILP FORMULATION

This section answers assignment Task 3.2. The full mathematical model is given once in Section 3; this task explains how it is used computationally.

A. MILP Formulation

The model is a mixed-integer linear program because all balances, colour constraints, quality constraints, and process-linking constraints are linear, while z_{malt} and z_{mash} are binary. The two binary variables determine whether fixed malting and mashing costs are charged.

B. LP Relaxation

The relaxed model uses the same equations as the MILP but replaces the binary restrictions by

$$0 \leq z_{\text{malt}}, z_{\text{mash}} \leq 1. \quad (37)$$

For cost minimisation, this gives a lower bound on the MILP optimum. A relaxed solution may contain fractional process variables, so it is not necessarily a valid production plan.

C. Brute-Force Method

Enumerating the binary variables is exact in this project because there are only two process choices and only three valid combinations:

$$(z_{\text{malt}}, z_{\text{mash}}) \in \{(0, 0), (0, 1), (1, 1)\}.$$

For each combination, the binary variables are fixed and the remaining continuous LP is solved. The best feasible solution among these LPs is the original MILP solution.

D. Programmatic Implementation

The implementation is written in Python. The helper file `brew_utils.py` reads `dataset_EN.txt`, defines the conversion constants, and provides calculation helpers. Each notebook builds the vector c , equality matrix A_{eq} , inequality matrix A_{ub} , right-hand sides, and variable bounds for `scipy.optimize.linprog`. More advanced LP or MILP solvers are not used. Result tables report only non-zero decision variables rather than the full optimisation vector.

6. TASK 3: 5000 LITRE AMBER BEER

A. Problem Setup

The requested production volume is 5000 L. The beer colour is amber, so the EBC bounds are

$$20 \leq \text{EBC} \leq 29.$$

No quality constraints are imposed in this task.

TABLE III
TASK 3 LP RELAXATION RESULT

Item	Result
Cost	4497.687
z_{malt}	0
z_{mash}	1
Achieved EBC	29.000

TABLE IV
TASK 3 NON-ZERO INGREDIENTS IN THE RELAXED SOLUTION

Group	Variable	Amount (kg)
Malt	malt04	685.349
Malt	malt06	268.598
Hops	hop07	6.842
Yeast	yeast05	3.750

TABLE V
TASK 3 ROUNDED-PROCESS RE-SOLVE

Item	Result
Rounded process choice	(0, 1)
Re-solved cost	4497.687
Achieved EBC	29.000
Ingredient mix	same as Table IV

B. LP Relaxation Result

The LP relaxation gives a cost of 4497.687. In this case the relaxed process variables are already integral: $z_{\text{malt}} = 0$ and $z_{\text{mash}} = 1$. The achieved colour is $\text{EBC} = 29.000$.

C. Why Most Ingredients Are Unused

Most ingredients are unused because an LP optimum usually occurs at a corner point of the feasible region. The solver selects the cheapest combination that satisfies the active volume and colour constraints. Ingredients that are more expensive or have less useful EBC values receive zero mass.

D. Rounded Relaxation

The rounding step is applied only to the binary process variables. The relaxed solution gives $(z_{\text{malt}}, z_{\text{mash}}) = (0, 1)$, so the rounded process choice is also (0, 1). The continuous LP is then re-solved with that process choice fixed. Directly rounding all ingredient quantities would be incorrect because ingredient amounts are continuous production decisions.

E. Original MILP Result

The brute-force MILP solve checks the valid process choices (0, 0), (0, 1), and (1, 1). The optimal process choice is (0, 1), with total cost 4497.687 and achieved $\text{EBC} = 29.000$.

TABLE VI
TASK 3 ORIGINAL MILP RESULT

Item	Result
Selected process choice	(0, 1)
Total cost	4497.687
Achieved EBC	29.000
Ingredient mix	same as Table IV



Fig. 2. Task 4 costs marked on the Task 1 cost-volume plot. Replace this placeholder with the plot exported from task_34.ipynb.

7. TASK 4: LESS OR MORE?

A. Results for 50 L and 25000 L

The amber-beer optimisation is repeated for 50 L, 5000 L, and 25000 L. The relaxation, rounded-process solution, and MILP solution coincide for these three cases because the relaxed process variables are already integral.

B. Marked Costs on Task 1 Plot

The three production volumes from Table VII are marked on the cost-volume plot from Task 1.

C. Cost per Litre Discussion

The cost per litre decreases as the production volume increases. The 50 L batch is expensive per litre because any fixed process cost is spread over very few litres. Larger batches spread fixed costs over more output, so the cost per litre is lower. The 25000 L case has the lowest cost per litre in Table VII.

8. TASK 5: FULL PROBLEM WITH QUALITY CONSTRAINTS

A. Problem Setup

This task uses the same production volume and colour as Task 3, but adds minimum average quality constraints. The setup is 5000 L amber beer with

$$Q_{\min}^S = 3, \quad Q_{\min}^H = 3, \quad Q_{\min}^Y = 3.$$

B. Optimal Solution

The optimal quality-constrained solution uses process choice $(z_{\text{malt}}, z_{\text{mash}}) = (0, 1)$. The total cost is 4891.923 and the achieved EBC is 20.000. All three achieved group qualities are 3.000.

TABLE VII
TASK 4 VOLUME COMPARISON

Beer volume (L)	Relaxation cost	Rounded cost	MILP cost	MILP cost per litre
50	67.767	67.767	67.767	1.355
5000	4497.687	4497.687	4497.687	0.900
25000	13420.213	13420.213	13420.213	0.537

TABLE VIII
TASK 5 PROBLEM SETUP

Parameter	Value
Beer volume	5000 L
Colour	amber
EBC bounds	20 to 29
Malt-source quality	at least 3
Hop quality	at least 3
Yeast quality	at least 3

TABLE IX
TASK 5 OPTIMAL SOLUTION SUMMARY

Item	Result
Total cost	4891.923
Process choice	(0, 1)
Achieved EBC	20.000
Malt-source quality	3.000
Hop quality	3.000
Yeast quality	3.000

TABLE X
TASK 5 NON-ZERO INGREDIENTS

Group	Variable	Amount (kg)
Malt	malt03	901.457
Malt	malt06	17.497
Malt	malt12	34.994
Hops	hop07	3.421
Hops	hop10	3.421
Yeast	yeast03	3.750

C. Comparison with Task 3

The quality constraints increase the cost from 4497.687 in Task 3 to 4891.923 in Task 5. This is expected because the feasible region becomes smaller. The model can no longer use only the cheapest low-quality combination from Task 3; it must choose ingredients whose weighted averages satisfy the three quality lower bounds. The optimal EBC also moves from the upper amber boundary in Task 3 to the lower amber boundary in Task 5.

9. TASK 6: FROM LIGHT BLOND TO BLACK

A. Results for All Beer Colours

The full quality-constrained model from Task 5 is solved for each colour class. Black beer is modelled as $EBC \geq 120 +$

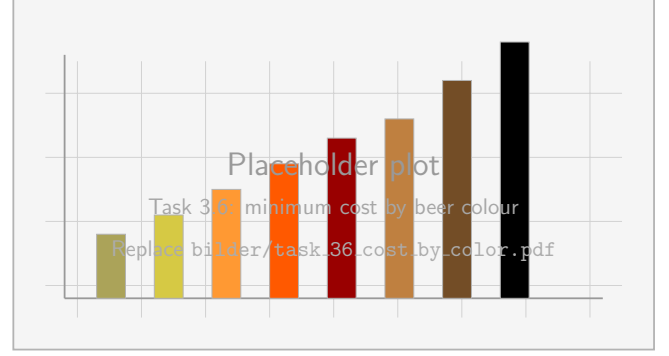


Fig. 3. Task 6 minimum cost by beer colour. Replace this placeholder with the plot exported from `task_36.ipynb`.

ϵ_{EBC} , with $\epsilon_{EBC} = 10^{-6}$, because the assignment specifies $EBC > 120$.

B. Cost as a Function of Beer Colour

The minimum cost for each feasible colour is plotted in Figure 3.

C. Explanation

The cost differences are caused by the available EBC values and quality scores in the dataset. Some colours can be reached with cheap ingredients while still satisfying the quality constraints. Other colours require a more expensive mix to move the weighted EBC average into the required interval. In this run, all feasible colour classes use mashing and avoid malting, so the main variation comes from the selected malt and extract sources rather than from different fixed process choices. Light blond is infeasible under the quality-constrained setup, while black is feasible only above the strict 120 EBC boundary represented by $120 + \epsilon_{EBC}$.

10. TASK 7: MAXIMISING THE QUALITY

A. New Objective

To maximise quality, introduce a scalar variable q representing the minimum achieved quality over the malt-source, hop, and yeast groups. The objective is

$$\max q, \quad (38)$$

or equivalently $\min -q$ when using `scipy.optimize.linprog`.

TABLE XI
TASK 6 RESULTS FOR ALL BEER COLOURS

Colour	EBC range	Feasible	Cost	Achieved EBC	Notes
light blond	6–8	no	–	–	No feasible quality-constrained mix found.
blond	9–12	yes	4919.601	12.000	Uses the upper blond boundary.
gold	13–19	yes	4883.000	14.535	Cheapest feasible colour in this run.
amber	20–29	yes	4891.923	20.000	Same setup as Task 5.
copper	30–45	yes	4908.251	30.000	Uses the lower colour boundary.
brown	46–75	yes	4934.376	46.000	Darker colour requires a different source mix.
dark brown	76–120	yes	4983.359	76.000	Upper bound remains 120.
black	> 120	yes	5061.830	120.000001	Modelled as $120 + \epsilon_{\text{EBC}}$.

TABLE XII
TASK 7 QUALITY MAXIMISATION RESULT

Item	Result
Maximum minimum quality	5.000
Quality 5 feasible?	yes
Process choice	(0, 1)
Achieved EBC	20.000
Malt-source quality	5.000
Hop quality	5.000
Yeast quality	5.000

TABLE XIII
TASK 7 NON-ZERO INGREDIENTS

Group	Variable	Amount (kg)
Malt	malt11	870.772
Malt extract	maltextract03	66.541
Hops	hop10	6.842
Yeast	yeast01	3.750

B. Quality Constraints

The volume, colour, and process constraints remain active. The three group-quality constraints are rewritten with q as the lower bound:

$$\sum_{i \in B} Q_i^B (0.6b_i) + \sum_{i \in M} Q_i^M (0.8m_i) + \sum_{i \in E} Q_i^E e_i \geq qM_{\text{tot}}, \quad (39)$$

$$\sum_{i \in H} Q_i^H h_i \geq q \sum_{i \in H} h_i, \quad (40)$$

$$\sum_{i \in Y} Q_i^Y y_i \geq q \sum_{i \in Y} y_i. \quad (41)$$

The bound $0 \leq q \leq 5$ is added. The binary process variables are still handled by the same brute-force enumeration as in Section 3.7.

C. Can the Minimum Quality Be 5?

Yes. The optimisation reaches $q^* = 5.000$, so minimum quality 5 is feasible. Since 5 is also the upper bound on q , this is the best possible value. No ingredient group prevents quality 5 in the computed solution: the malt-source, hop, and yeast qualities all reach 5 while satisfying the amber colour constraint.

11. CONCLUSION

The Brewery Plat-Eau tasks are solved with one common linear optimisation model and task-specific parameter choices. Fixed malting and mashing costs are represented with two binary variables. The LP relaxation gives a lower bound, while brute-force enumeration over the valid binary process choices gives the original MILP solution using only `scipy.optimize.linprog`.

The results show how volume, colour, and quality constraints change the cheapest production plan. Fixed process costs explain why small batches are more expensive per litre, and the EBC and quality constraints explain why some colours or ingredient mixes cost more than others.

APPENDIX A FINAL INSERTION NOTES

The main task answers are placed in Sections 4–10. The remaining manual report work is to replace the placeholder figure files in `bilder/` with the final exported plots or screenshots from the notebooks. The numerical tables in the main text should be checked against a clean run of `task_31.ipynb` through `task_37.ipynb` before submission.

REFERENCES

- [1] “Brewery Plat-Eau,” Optimisation Techniques project assignment, `Project_EN_26.pdf`, project assignment.

- [2] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, Í. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, and P. van Mulbregt, “SciPy 1.0: Fundamental algorithms for scientific computing in python,” *Nature Methods*, vol. 17, pp. 261–272, 2020.